SC3102: Signals, Systems, and Transforms

Tutorial 05: Differential equations

All signals in the questions are real and continuous-time.

1. Consider the differential equation

$$y'(t) + 2y(t) = 5\cos(t)u(t), \qquad t > 0,$$

with initial value $y(0^-) = 3$.

(a) Use the Laplace transform to find the complete responses, y for $t > 0$.

(b) Identify the transient and steady-state components.

(c) Find the initial value that eliminates the transient.

2. Consider the differential equation

$$y''(t) + 3y'(t) + 2y(t) = 10u(t), \qquad t > 0,$$

with initial conditions $y(0^-) = 1$ and $y'(0^-) = -1$.

(a) Use the Laplace transform to find the zero-input, zero-state and complete responses.

(b) Does the final value theorem apply here? If so, use it to identify the constant steady-state behavior of the output signal and justify its application.

3. Suppose a continuous-time LTI system satisfies

$$y''(t) + 4y'(t) + 3y(t) = 2x'(t) + 4x(t), \qquad t \in \mathbb{R}.$$

(a) Use the bilateral Laplace transform to find the system function:

$$H(s) = \frac{Y(s)}{X(s)}$$

and make its pole-zero plot.

(b) Sketch all possible ROCs of $H(s)$. Which of these systems are causal and which are BIBO stable? Are any of them both?

(c) Find the causal and BIBO-stable impulse responses.

Useful Properties and Inverse Pairs

Unilateral Laplace Transform Pairs:

Signal $x(t)$	Transform $X_+(s)$	ROC
1	$\frac{1}{s}$	$\Re s > 0$
e^{pt}	$\frac{1}{s-p}$	$\Re s > \Re p$
te^{pt}	$\frac{1}{(s-p)^2}$	$\Re s > \Re p$
$\cos(\omega_0 t)$	$\frac{s}{s^2+\omega_0^2}$	$\Re s > 0$
$\sin(\omega_0 t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$\Re s > 0$
$e^{\alpha t}\cos(\omega_0 t)$	$\frac{s-\alpha}{(s-\alpha)^2+\omega_0^2}$	$\Re s > \alpha$
$e^{\alpha t}\sin(\omega_0 t)$	$\frac{\omega_0}{(s-\alpha)^2+\omega_0^2}$	$\Re s > \alpha$

Bilateral Laplace Transform Pairs:

Signal $x(t)$	Transform $X(s)$	ROC
$e^{pt}u(t)$	$\frac{1}{s-p}$	$\Re s > \Re p$
$-e^{pt}u(-t)$	$\frac{1}{s-p}$	$\Re s < \Re p$
$te^{pt}u(t)$	$\frac{1}{(s-p)^2}$	$\Re s > \Re p$
$\cos(\omega_0 t)u(t)$	$\frac{s}{s^2+\omega_0^2}$	$\Re s > 0$
$\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$\Re s > 0$
$e^{\alpha t}\cos(\omega_0 t)u(t)$	$\frac{s-\alpha}{(s-\alpha)^2+\omega_0^2}$	$\Re s > \alpha$
$e^{\alpha t}\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{(s-\alpha)^2+\omega_0^2}$	$\Re s > \alpha$

Unilateral Laplace Transform Differentiation Rules:

$$\mathcal{L}_+\{x'(t)\} = sX_+(s) - x(0^-),$$
$$\mathcal{L}_+\{x''(t)\} = s^2X_+(s) - sx(0^-) - x'(0^-).$$

Bilateral Laplace Transform Differentiation Rules:

$$\mathcal{L}\{x'(t)\} = sX(s), \qquad \mathcal{L}\{x''(t)\} = s^2X(s).$$